

Coppervale Semiconductor Fab

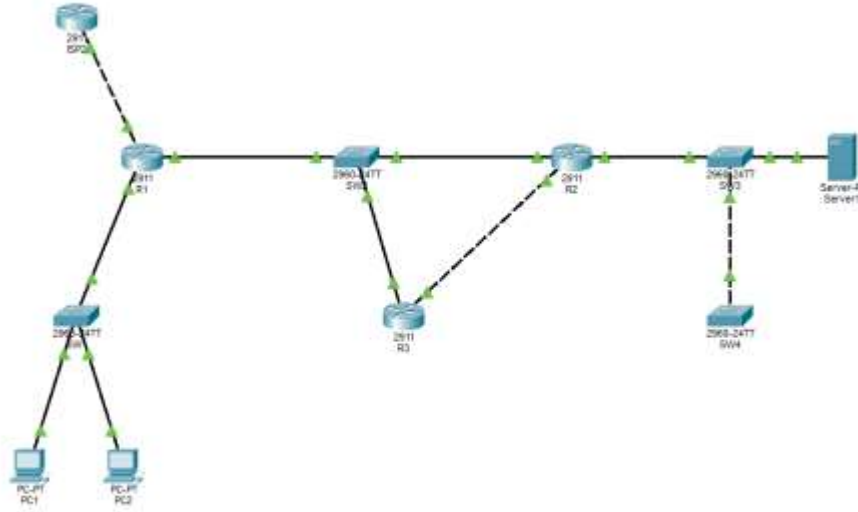
Network Outage Investigation: Capstone Diagnostics

Five symptoms. Root causes, fixes and verification across STP, OSPF, EIGRP, BGP and ICMP.

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Scenario and lab topology



Built and tested in Cisco Packet Tracer (Cisco 2911 routers, 2960 switches).

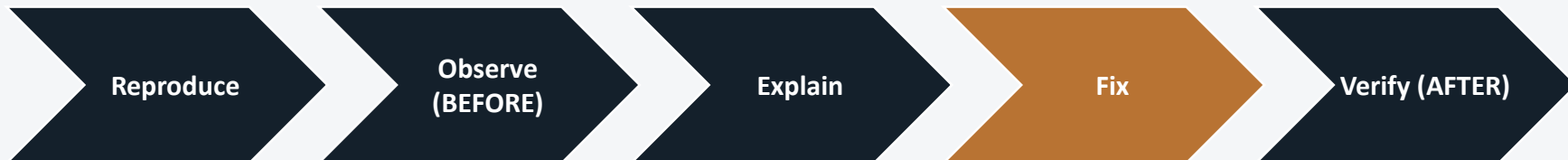
Three sites plus a backup ISP

- HQ: R1, OSPF hub and eBGP edge
- Fab Plant: R2 (OSPF + EIGRP), SW3, Server1
- Test & Packaging: R3, EIGRP stub
- ISP2 (AS 65002) advertises 5 test prefixes

Where each fault lives

- A SW3/SW4 (STP)
- B R1/R2 WAN (OSPF)
- C R3 (EIGRP)
- D R1/ISP2 (BGP)
- E PC2 to Server1 (ICMP)

Troubleshooting method



Healthy baseline first

- show ip ospf neighbor: R2 and R3 FULL
- show ip bgp summary: 5 prefixes from ISP2
- show spanning-tree inconsistentports: none
- PC1 pings Server1 and 192.168.30.1

Rules followed

- Break and fix one symptom at a time
- Same show command before and after
- Change only the line that caused the fault
- Keep the protections (Root Guard, max-prefix) on



Root Guard blocks a Fab access port

STP | SW3 / SW4 | Objective 1

Symptom

SW3 port Fa0/3 is blocked with state root-inconsistent.

Root cause

SW4 priority was lowered to 0, so it sends a superior BPDU toward SW3.

Why it blocks

Root Guard stops any switch from becoming root, because the root decides the whole Layer 2 tree. The port blocks instead of accepting the BPDU. The protection worked; the priority was the error.

Fix

Restore SW4 priority to 32768. Keep Root Guard on.

BEFORE

```
SW3# show spanning-tree inconsistentports
VLAN0001 Fa0/3 Root Inconsistent
%SPANTREE-2-ROOTGUARD_BLOCK: Root guard
blocking port Fa0/3 on VLAN0001
```

FIX

```
SW4(config)# spanning-tree vlan 1 priority 32768
```

AFTER

```
SW3# show spanning-tree inconsistentports
Number of inconsistent ports ... : 0
%SPANTREE-2-ROOTGUARD_UNBLOCK
```



OSPF neighbor stuck in ATTEMPT

OSPF | HQ to Fab multipoint WAN | Objective 2

Symptom

R2 stays in ATTEMPT on the shared WAN segment while R3 reaches FULL.

Root cause

R2 Gi0/0 was reset to BROADCAST, but hub R1 is NON_BROADCAST. R1 sends unicast hellos and R2 does not answer them.

Why ATTEMPT

On NBMA there is no multicast discovery. The hub needs a neighbor statement and the spoke must use the same network type. If the statement is missing, R2 never appears at all.

Fix

R2 back to non-broadcast, priority 0. Confirm neighbor 10.0.0.2 on R1.

BEFORE

```
R1# show ip ospf neighbor
0.0.0.0    0 ATTEMPT/DROTHER 10.0.0.2
3.3.3.3    0 FULL/DROTHER    10.0.0.3
R1 Gi0/1:  NON_BROADCAST, hello 30
R2 Gi0/0:  BROADCAST, hello 10
```

FIX

```
R2(config-if)# ip ospf network non-broadcast
R2(config-if)# ip ospf priority 0
R1(config-router)# neighbor 10.0.0.2
```

AFTER

```
2.2.2.2    0 FULL/DROTHER    10.0.0.2
3.3.3.3    0 FULL/DROTHER    10.0.0.3
```



EIGRP stub hides the Test & Packaging route

EIGRP | R3 to R2 | Objective 3

Symptom

192.168.30.0/24 is in R3's own table but missing on R2.
Ping from R1 fails.

Root cause

R3 is set to eigrp stub receive-only.

Stub types

Default / connected summary advertises connected and summary routes. Receive-only advertises nothing, not even its own LAN. A wrong type silently withholds a route.

Fix

Use eigrp stub connected summary.

BEFORE

```
R3# show ip route 192.168.30.0
C 192.168.30.0/24 is directly connected
R2# show ip route eigrp (route absent)
R3# show ip protocols -> Stub, receive-only
R1# ping 192.168.30.1 -> timed out
```

FIX

```
R3(config-router)# eigrp stub connected summary
```

AFTER

```
D 192.168.30.0/24 [90/130816] via 10.2.0.2
R1# ping 192.168.30.1 -> 100 percent
```



eBGP session to ISP2 keeps resetting

BGP | HQ Internet edge | Objective 4

Symptom

The session to the secondary ISP resets again and again.

Root cause

R1 has maximum-prefix 3, but ISP2 advertises 5 prefixes.

Why tear down

The limit protects router memory and CPU from a route leak or attack. Dropping the extra routes silently would leave an incomplete table, so BGP fails loudly.

Safe fix

Confirm with the ISP that the 5 prefixes are legitimate, raise the limit with headroom and a warning, then clear the session. Never remove the limit.

BEFORE

```
%BGP-3-MAXPFX: prefix received from  
172.16.2.2 reaches 4, max 3  
%BGP-5-ADJCHANGE: neighbor 172.16.2.2  
Down - Peer over limit  
172.16.2.2 4 65002 ... Idle (PfxCt)
```

FIX

```
neighbor 172.16.2.2 maximum-prefix 10 80  
R1# clear ip bgp 172.16.2.2
```

AFTER

```
172.16.2.2 4 65002 ... 00:01:10 5  
R1# show ip route bgp -> five B 198.51.10x.0/24
```



Burst of fragmented ICMP echo

ICMP | Fab access layer capture | Objective 5

Pattern

Large echo requests above the 1500-byte MTU, so they fragment. Same source 192.168.10.66, same target 192.168.20.100, hundreds in seconds.

Likely intent

A ping flood or Ping-of-Death style DoS to exhaust CPU, fragment-reassembly memory or bandwidth on the Fab server.

Mitigation 1

ACL on R2 Gi0/0 inbound: block the attacker, allow replies and other traffic.

Mitigation 2

Rate-limit ICMP echo (police 64000) and filter fragments at the Fab boundary. Shown as representative; Packet Tracer cannot verify it.

CAPTURE

```
icmp.type == 8 && ip.flags.mf == 1  
PC2> ping -l 10000 -n 200 192.168.20.100
```

FIX

```
ip access-list extended PROTECT-FAB  
deny icmp host 192.168.10.66  
host 192.168.20.100 echo  
permit icmp any any echo-reply  
permit ip any any  
interface gi0/0 / ip access-group PROTECT-FAB in
```

AFTER

```
10 deny icmp host ... echo (45 matches)  
PC2 ping timed out; PC1 still works
```

Root-cause analysis

Sym.	Likely cause	Evidence	Fix	Verification
A	SW4 priority 0 sends a superior BPDU	ROOTGUARD_BLOCK log; Fa0/3 Root Inconsistent	Set SW4 priority to 32768	inconsistentports = 0; UNBLOCK log
B	R2 not NON_BROADCAST; hub hellos unanswered	R2 in ATTEMPT, R3 FULL; types differ	Non-broadcast on R2; neighbor on R1	R2 and R3 both FULL
C	R3 uses eigrp stub receive-only	Route on R3 only; show ip protocols: receive-only	eigrp stub connected summary	D route on R2; ping works
D	Max-prefix 3, ISP2 sends 5	MAXPFX log; Idle (PfxCt)	maximum-prefix 10 80; clear ip bgp	Established, 5 prefixes
E	Fragmented echo flood from 192.168.10.66	MF set; same src and dst; rapid burst	ACL on R2; ICMP rate limit	ACL counters; PC1 still pings

Are the five symptoms related?

Possible shared change window



Both sit in the Fab / Test & Packaging area and both were changed "recently". Check the change log, but they remain two misconfigurations with two fixes.

Independent faults



B is the WAN link, D is only on the HQ Internet edge, and E comes from a host, not a router setting.

Conclusion: more than one unrelated fault.

No common fingerprint (such as every protocol failing on one router). Root Guard, max-prefix and the stub all did their job; the causes are upstream. Each fix must be verified on its own.

Prioritized remediation plan

1	B OSPF WAN	HQ to Fab backbone degraded; ERP and production depend on it	<code>show ip ospf neighbor</code>
2	C EIGRP stub	Test & Packaging unreachable from the rest of the company	<code>ping 192.168.30.1</code>
3	A Root Guard	Devices behind the blocked Fab port lose network	<code>show spanning-tree inconsistentports</code>
4	E ICMP flood	Slowdown risk on a Fab host; no outage yet	<code>show access-lists</code>
5	D BGP secondary	Backup path only; primary ISP still carries traffic	<code>show ip bgp summary</code>

Quick win: the E ACL takes about 2 minutes, so apply it first as containment while B, C and A are fixed.

Prevention and honest limits

Prevent repeats

- Change control and change-log review
- Root Guard plus BPDU Guard on edge ports
- Prefix limits with headroom and warnings
- Stub-design review for EIGRP
- ICMP rate limiting and ACLs at the Fab boundary

Lab limits

- Command outputs here are representative, taken from the lab guide, not captured from my own run
- Packet Tracer has no Wireshark; Simulation mode stands in
- ICMP policing is shown as concept only; the ACL is the verified mitigation
- Project file: coppervale_phase1.pkt